COURSE CODE/TITLE:	CPE 031 - VISUALIZATIONS AND DATA ANALYSIS	**SECTION:**	CPE21S1
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ACTIVITY NO. 13 [Designing a Purpose-Driven Data Dashboard]

1. PROCEDURE OUTPUT

Task 1:
Define the Purpose

Mendoza, Roy Gabriel

This academic dashboard is designed to monitor student progress and academic standing across multiple subjects through interactive visualizations. By using metrics for test scores and attendance percentages, it allows the instructors and department heads to easily identify the gaps and the underperforming students. Lastly, this dashboard shows the assessment data and actionable classroom intervention strategies to improve overall student retention and academic performance

Mendoza, Ryo Shin

The purpose of my dashboard is to analyze and visualize student performance and attendance across five subjects. It presents the average scores, the relationship between attendance and scores, the distribution of scores, and the average attendance per subject. These visualizations make it easier to compare student performance and identify patterns in the data. Overall, the dashboard helps present information clearly and supports a better understanding of students' academic performance.

Obias, Jave Joaquin

The purpose of the dashboard is to evaluate academic progress, also this organizes individual student grades and attendance records into a single visual view. It allows to examine subject performance, spot relationships between attendance and overall marks, and filter information by subject. Overall, this dashboard makes student evaluation more efficient and helps guide timely interventions to support student success.

Task 2:
Design for User Needs

Code

Teacher Dashboard

```
# Set performance thresholds
score_threshold = 75
attendance_threshold = 80

# Summarize each student's performance
student_summary = df.groupby("Student", as_index=False).agg(
    Average_Score=("Score", "mean"),
    Average_Attendance=("Attendance (%)", "mean")
)

# Identify students who may need support
student_summary["Status"] = student_summary.apply(
    lambda row: "Needs Support"
    if row["Average_Score"] < score_threshold
    or row["Average_Attendance"] < attendance_threshold
    else "Satisfactory",
    axis=1
)

# Subject-level averages
subject_summary = df.groupby("Subject", as_index=False)["Score"].mean()

# Create dashboard
fig = make_subplots(
    rows=2, cols=2,
    subplot_titles=(
        "Average Score per Subject",
        "Student Attendance vs. Score",
        "Students Who Need Support",
        "Student Performance Summary"
    ),
    specs=[
        [{"type": "bar"}, {"type": "scatter"}],
        [{"type": "bar"}, {"type": "table"}]
    ],
    vertical_spacing=0.15,
    horizontal_spacing=0.10
)

# 1. Average scores by subject
fig.add_trace(
    go.Bar(
        x=subject_summary["Subject"],
```

```
            y=subject_summary["Score"],
            name="Average Score"
        ),
        row=1, col=1
    )

    # 2. Attendance vs. score
    fig.add_trace(
        go.Scatter(
            x=df["Attendance (%)"],
            y=df["Score"],
            mode="markers",
            text=df["Student"],
            name="Student Records",
            marker=dict(color="royalblue")
        ),
        row=1, col=2
    )

    # 3. Students needing support
    needs_support = student_summary[
        student_summary["Status"] == "Needs Support"
    ].sort_values("Average_Score")

    fig.add_trace(
        go.Bar(
            x=needs_support["Student"],
            y=needs_support["Average_Score"],
            name="Needs Support",
            marker_color="indianred"
        ),
        row=2, col=1
    )

    # 4. Student summary table
    fig.add_trace(
        go.Table(
            header=dict(
                values=["Student", "Average Score",
                        "Average Attendance", "Status"]
            ),
            cells=dict(
                values=[
                    student_summary["Student"],
                    student_summary["Average_Score"].round(1),
                    student_summary["Status"]
                ]
            )
        ),
        row=2, col=2
    )

    fig.update_layout(
        title="Teacher Dashboard: Student Performance Monitoring",
        height=850,
        width=1200,
        showlegend=False
    )

    fig.update_yaxes(title_text="Average Score", row=1, col=1)
    fig.update_xaxes(title_text="Attendance (%)", row=1, col=2)
    fig.update_yaxes(title_text="Score", row=1, col=2)
    fig.update_yaxes(title_text="Average Score", row=2, col=1)

    fig.show()
```

Output



Mendoza, Roy	My task 2 is tailored for the requirement of a department head by focusing exclusively on subject level averages. By organizing the visualization into a clear 1 row 2 column subplot structure, this allows the head to evaluate both the mean scores and the mean attendance side by side in an instant glance. Additionally, adding precise data labels onto the bar chart eliminates the need for manual estimations, ensuring clarity. Lastly, this summary empowers the efficiency of the head and the leadership to assess the overall department performance and make informed academic decisions.
Mendoza, Ryo	In Task 2, we modified the dashboard to meet the needs of teachers by highlighting students who may be underperforming based on their grades and attendance. The bar chart and summary table make it easier to identify students who need additional academic support. This activity helped me understand how dashboards can be designed for specific users and how data visualization can help teachers make informed decisions to improve student performance.
Obias, Jave	In this task 2, we develop the dashboard for the requirements needed by the teachers that highlights the grades and the attendance of the students. Using bar charts, a scatter plot, and a summary table, it makes it much easier to track individual student performance, spot who needs extra academic support, and monitor overall class trends at a glance.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary:

Design Your Own Dashboard: “Smart Campus Energy Monitor” Create a dashboard that shows the energy usage of different campus areas (e.g., laboratories, offices, classrooms) over time.

1. Simulate or use sample data (Energy consumption per hour or per day).
2. Include at least three charts (bar, line, pie).
3. Ensure your design reflects a clear purpose and target user (e.g., facilities manager).
4. Write a short explanation of your engineering decisions (why you chose certain colors, layouts, or interactivity).

Code

```
# Generate sample hourly energy consumption
np.random.seed(10)

hours = range(24)
areas = ["Laboratories", "Classrooms", "Offices", "Library"]

base_usage = {
    "Laboratories": 18,
    "Classrooms": 12,
    "Offices": 8,
    "Library": 10
}

records = []

for hour in hours:
    for area in areas:
        if 7 <= hour <= 18:
            usage = base_usage[area] * np.random.uniform(0.8, 1.4)
        else:
            usage = base_usage[area] * np.random.uniform(0.1, 0.5)

        records.append({
            "Hour": hour,
            "Area": area,
            "Energy (kWh)": round(usage, 2)
        })

df_energy = pd.DataFrame(records)

# Summarize energy usage
area_usage = df_energy.groupby(
    "Area", as_index=False
)["Energy (kWh)"].sum()

hourly_usage = df_energy.groupby(
    "Hour", as_index=False
)["Energy (kWh)"].sum()

display(df_energy.head())
```

```python
fig1 = px.bar(
    area_usage,
    x="Area",
    y="Energy (kWh)",
    title="Total Energy Consumption by Campus Area",
    color="Area",
    color_discrete_sequence=[
        "#2563EB", "#14B8A6", "#F59E0B", "#8B5CF6"
    ],
    text_auto=".2f"
)

fig1.update_layout(
    template="plotly_white",
    xaxis_title="Campus Area",
    yaxis_title="Total Energy Consumption (kWh)",
    showlegend=False
)

fig1.show()
```

```python
fig2 = px.line(
    hourly_usage,
    x="Hour",
    y="Energy (kWh)",
    title="Hourly Campus Energy Consumption",
    markers=True
)

fig2.update_traces(
    line=dict(color="#2563EB", width=3)
)

fig2.update_layout(
    template="plotly_white",
    xaxis_title="Hour of the Day",
    yaxis_title="Total Energy Consumption (kWh)",
    xaxis=dict(dtick=2)
)

fig2.show()
```

```python
fig3 = px.pie(
    area_usage,
    names="Area",
    values="Energy (kWh)",
    title="Share of Total Energy Consumption by Campus Area",
    hole=0.35,
    color_discrete_sequence=[
        "#2563EB", "#14B8A6", "#F59E0B", "#8B5CF6"
    ]
)

fig3.update_traces(
    textinfo="label+percent",
    hovertemplate=(
        "%{label}<br>"
        "Energy: %{value:.2f} kWh<br>"
        "Share: %{percent}<extra></extra>"
    )
)

fig3.update_layout(
    template="plotly_white"
)

fig3.show()
```

```python
total_energy = df_energy["Energy (kWh)"].sum()

print(
    f"Total Campus Energy Consumption: "
    f"{total_energy:.2f} kWh"
)

print("\nEnergy Consumption by Campus Area:")
display(area_usage.round(2))
```

```
Total Campus Energy Consumption: 796.76 kWh

Energy Consumption by Campus Area:
```

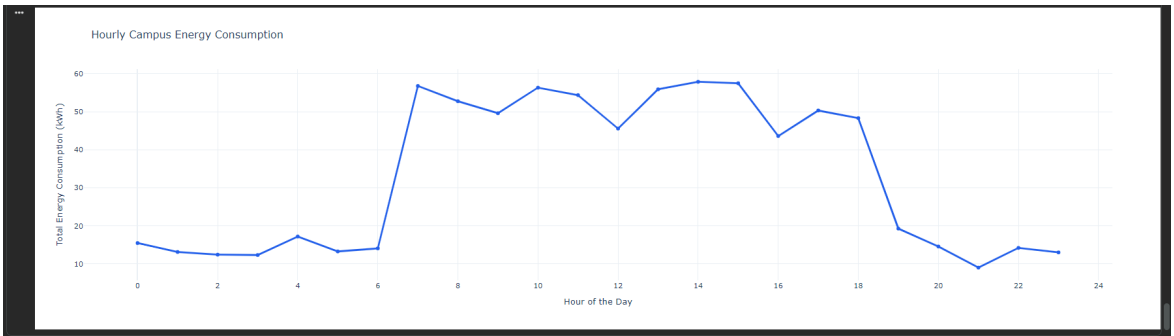
	Area	Energy (kWh)
0	Classrooms	200.19
1	Laboratories	299.09
2	Library	169.83
3	Offices	127.65

Output



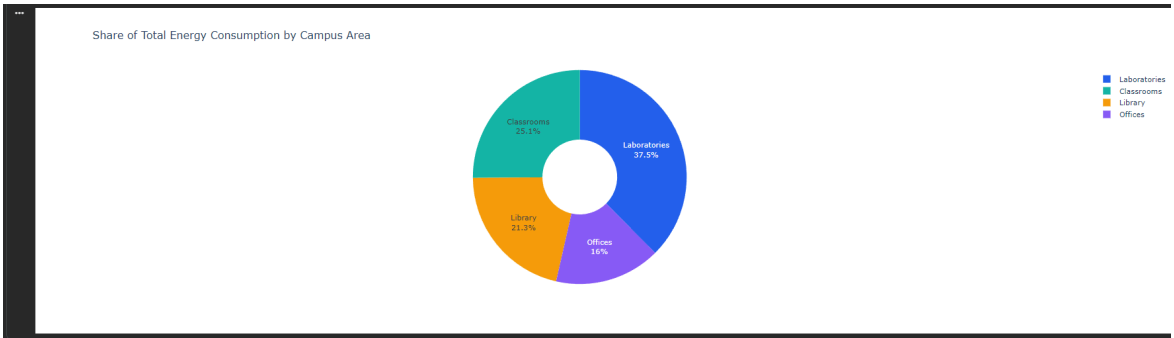
Bar Chart — Total Energy Consumption by Campus Area

The bar chart compares the total energy consumption of laboratories, classrooms, offices, and the library. It helps the facilities manager identify which campus area uses the most electricity and determine where energy-saving measures may be needed.



Line Chart — Hourly Campus Energy Consumption

The line chart shows how the campus's total energy consumption changes throughout the 24-hour period. It helps the facilities manager identify peak usage hours and understand when electricity demand is highest or lowest.



Pie Chart — Share of Total Energy Consumption

The pie chart shows the percentage of total energy consumption contributed by each campus area. It makes it easier to compare each area's share of electricity use and determine which areas contribute most to overall consumption.

Analysis

Mendoza, Roy Gabriel	For the Smart Campus Energy Monitor Dashboard, the design decisions that are made to make sure of clarity and usability to almost all of the management. First, a multi chart layout with a bar chart for totals, a line chart for the trends overtime, and a pie chart for a proportional shares, was selected to allow the involved parties to see the macro level totals, peak load hours and the resource distribution without too much visual cluttering. Second, high contrast and distinct categorical color palettes were used so that all different rooms and facilities are distinguishible at first glance.

	Finally, using plotly's interactive tools, allowed us to get the data into the exact kilowatt hour values and time usage which the involved parties can use for effective energy conservation planning.
Mendoza, Ryo Shin	In this activity, we created a Smart Campus Energy Monitor dashboard designed for facilities managers to monitor electricity usage in different campus areas. The bar chart compares total energy consumption, the line chart shows changes in energy usage over time, and the pie chart displays each area's share of total consumption. We used different colors to easily distinguish the areas and organized the charts for better readability. This dashboard helps facilities managers identify areas with high energy consumption and find opportunities to save electricity.
Obias, Jave Joaquin	In this supplementary part of the activity, we put together an energy monitoring dashboard to help facilities managers easily keep track of electricity use across different campus zones. We organized the layout using bar, line, and pie charts so that overall totals, hourly trends, and percentage shares could be checked all in one clean view without clutter. Adding distinct colors and interactive hover details also made it simple to spot heavy energy areas and plan out practical power-saving steps.

Conclusion	
Mendoza, Roy Gabriel	In conclusion of this activity, the process of designing and building a purposeful user centered data dashboards, using python and the power of plotly was successfully demonstrated. Task 1 made us make an effective dashboard that is a structured storytelling medium aligned with specific monitoring goals, while task 2 is tailored for the clean and organized subject level average scores and attendance into side by side sub plots with precise data labels. Finally the supplementary activity made us to make a clean and overall implementation on what we have learned in the procedure in this activity and the previous lessons, to maintain clarity, not to be overdosed with information, and the visual hierarchy, ensures that the dataset, whatever the size of them, are successfully converted into an insightful and effective storytelling material.
Mendoza, Ryo Shin	Through this activity, I learned how to use Python, Pandas, NumPy, and Plotly to create an interactive dashboard for monitoring campus energy consumption. The bar chart helped compare electricity usage across different campus areas, while the line chart showed changes in energy consumption throughout the day. The pie chart showed each area's share of total energy usage. I learned that choosing appropriate charts, colors, and layouts makes data easier to understand and interpret. Overall, the Smart Campus Energy Monitor demonstrates how data visualization can help facilities managers identify high energy consumption areas and make better decisions about saving electricity.
Obias, Jave Joaquin	In conclusion, working on this energy monitoring dashboard using Python,

	Pandas, and Plotly was a great way to put data visualization into practice. Using bar, line, and pie charts made it easy to compare usage, track hourly changes, and show percentage shares so facilities managers can easily spot high energy areas. Overall, this activity tied everything together nicely; Task 1 taught me how to treat a dashboard like a structured story with a clear purpose, while the other tasks helped me refine clean subplots, colors, and data labels.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Purpose of Use: **None**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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